

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define Mechatronics with its merits and demerits. Brief in detail any one of industrial application of Mechatronics.	CO1	6
B) Explain briefly any two of the following transducers: a. Linear-variable-differential transformer (LVDT) b. Optical Encoder c. Strain Gauges	CO1	6
C) Explain Analog to Digital Conversion and Describe any one method of analog-to-digital conversion with neat block diagram.	CO2	6
Q.2 Solve Any Two of the following.		12
A) What is the need for signal conditioning? Explain the steps involved in signal conditioning.	CO2	6
B) Explain with neat diagram DC Motor with its industrial applications in Mechatronic domain.	CO4	6
C) Define Transfer Function and Determine the transfer function $C(s) / R(s)$ of the system shown in Fig.1. by block diagram reduction method.	CO4	6

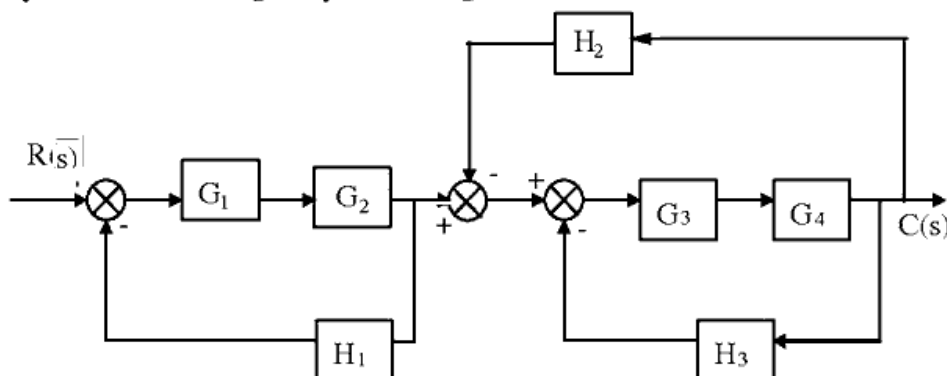


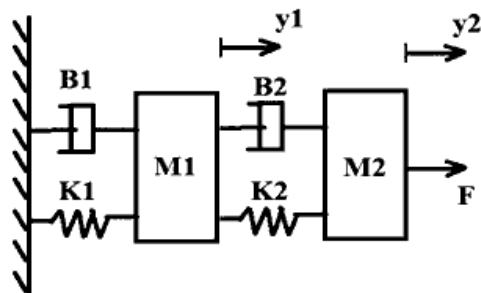
Fig.1.

Q.3 Solve Any Two of the following.		12
A) Explain pneumatic sequencing circuit diagram for 2 cylinders with Cascade method.	CO3	6

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| B) | Explain the different types of directional control valves with neat sketches. | CO3 | 6 |
| C) | What is a "Microprocessor"? Explain briefly 8085 microprocessor with the help of a block diagram and also uses of microprocessors? | CO5 | 6 |

Q.4 Solve Any Two of the following. **12**

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| A) | Explain Closed Loop Control System with block Diagram. Describe advantages and Disadvantages over Open Loop Control System. | CO3 | 6 |
| B) | Write the differential equations governing the mechanical translational system as shown in figure and determine the transfer function. | CO5 | 6 |



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| C) | Explain with block Diagram Programmable Logic Controller (PLC). Write any one industrial application using PLC Programming. | CO4 | 6 |
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Q. 5 Solve Any Two of the following. **12**

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| A) | Explain different control actions used in Closed Loop Control System (PID, PI, PD) and write PID Controllers in brief with advantages. | CO6 | 6 |
| B) | Explain the various Force Sensors with industrial applications. | CO1 | 6 |
| C) | What do you mean by bode plot? Sketch the bode plot for the following transfer function and determine the phase margin & gain margin. $G(S) = \frac{20}{s(1+3s)(1+4s)}$ | CO6 | 6 |

*** End ***